SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 4

Managerial Accounting & Costing Systems (Job-Order & Process Costing)

From accounting for outsiders to accounting for managers: classifying costs, cost behavior, and two product-costing systems — job-order (unique orders) and process costing (continuous mass production). Includes a full worked solution to the Nusantara CraftWorks case.

Material

1 Cost classification depends on purpose

Classifications serve different purposes: direct vs indirect (tracing), DM/DL/MOH (manufacturing), product vs period (statements), variable/fixed/mixed (behavior), differential/opportunity/sunk (decisions).

`Prime = DM + DL` `Conversion = DL + MOH` Selling and administrative costs are period costs; product costs wait in inventory until sale.

2 Cost behavior and decision costs

Type	Total	Per unit
Variable	Proportional	Constant
Fixed	Constant in relevant range	Falls as activity rises
Mixed	Y = a + bX	Depends

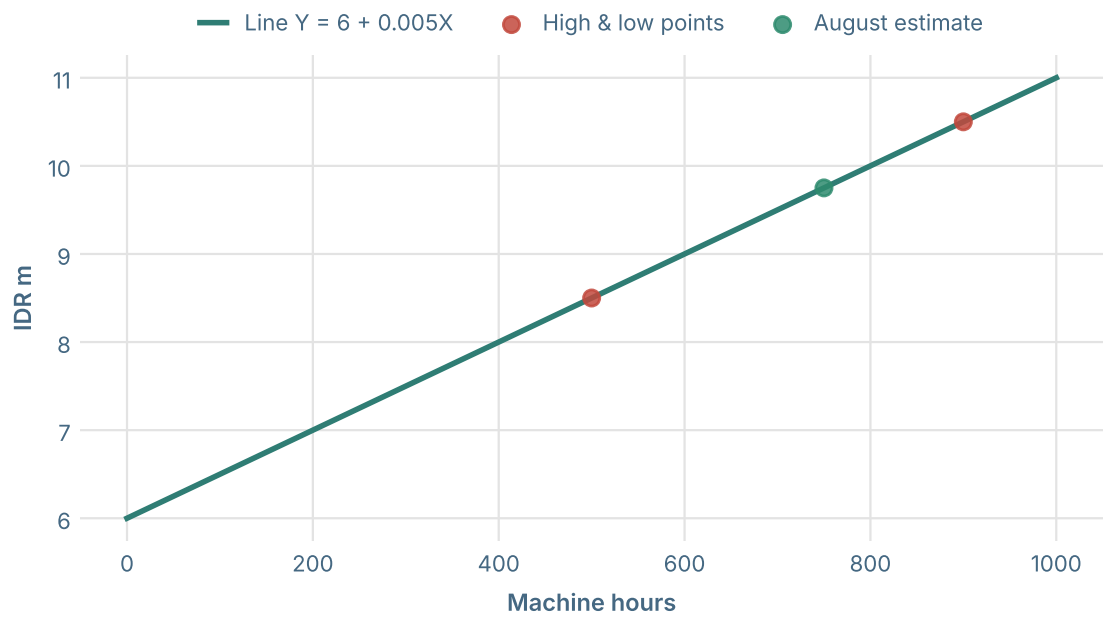
Cost drivers cause variable costs. Separate mixed costs by scattergraph, high-low (b = Δcost ÷ Δactivity) or regression. Decision costs: differential and opportunity costs are relevant; **sunk costs are ignored**.

3 High-low drawn: separating fixed and variable cost

Deep dive

High-low uses the highest and lowest activity points to split mixed costs.

Power cost vs machine hours (high-low)



Intercept = fixed cost; slope = variable cost per hour.

1. b = (10.5m − 8.5m) ÷ 400 = IDR 5,000/hour.
2. a = IDR 6,000,000.
3. August (750 h) = IDR 9,750,000.

Weakness

Only two points; regression uses all.

4 Job-order costing: for unique orders

Used for unique orders (aircraft, construction, consulting). The **job** is the cost object; the **job cost sheet** collects DM (requisitions), DL (time tickets) and applied overhead.

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POHR = Estimated overhead ÷ Estimated allocation base; Applied =
POHR × actual base
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Example: $640k ÷ 160k DLH = $4/DLH; 8 DLH → $32. Average unit cost ($118) isn't incremental cost because it contains fixed overhead.

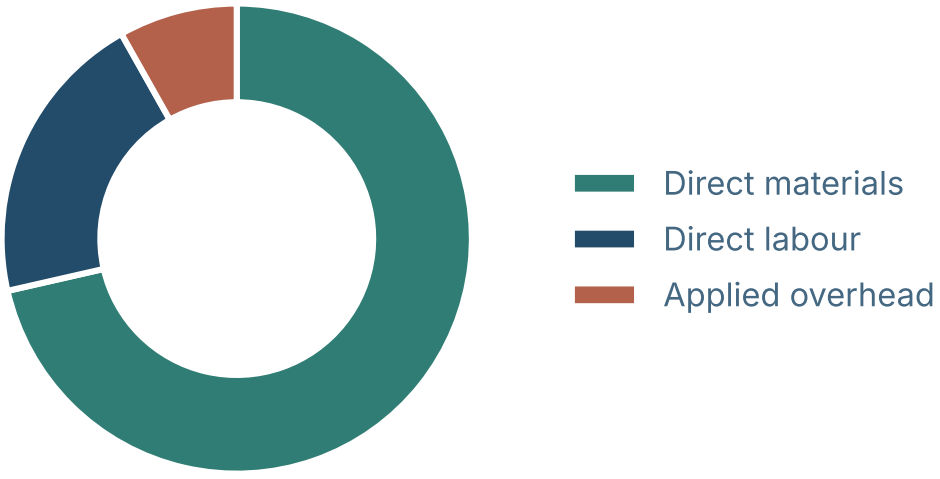
Flow: RM → WIP → FG (COGM) → COGS. COGM = total manufacturing costs + BWIP − EWIP; COGS = BFG + COGM − EFG ± under/overapplied overhead.

5 Job KR-118 cost and the job-order cost flow

Deep dive

Raw materials → WIP → Finished goods → COGS

Job KR-118 cost mix (IDR 58.8m)



IDR 58.8m	IDR 2.94m	~IDR 2.82m
Job total (20 chairs)	Average per chair	Incremental chair

6 Process costing: for uniform mass production

For homogeneous continuous production (beverages, cement, food). Costs accumulate by **department**; the production report is the key document.

Equivalent units: two half-done units = one complete unit. Weighted-average: EU = units transferred + EU in ending WIP; cost per EU = (BWIP cost + added cost) ÷ EU — separately for materials and conversion. Then cost ending WIP and transfers, and reconcile.

Operation costing is the hybrid: common processing plus customized finishing.

7 Equivalent units: why half-finished goods count partly Deep dive

WIP counts partly: 500 units at 40% materials and 20% conversion = 200 and 100 equivalent units.

Equivalent units: materials vs conversion




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Transferred 142.8m + ending WIP 3.3m = 146.1m ✔
```

Key takeaways

1. Cost classification depends on purpose.
2. Product costs → inventory → COGS; period costs expensed immediately.
3. Mixed cost Y = a + bX; separate via high-low or regression.
4. Ignore sunk costs; opportunity costs are relevant.
5. Job-order: cost per job; overhead applied via POHR.
6. Process: cost per department; equivalent units; reconcile.
7. Average unit cost ≠ incremental cost.

Case studies

Case & worked answers

Slide exercises — answers

MC1: wood direct/product; supervisor manufacturing overhead; commission period; head-office accounting administrative; factory depreciation indirect. MC2: A job-order, B process, C operation costing. MC3: POHR $4/MH; Job X $140; actual overhead isn't known in time. MC4: materials EU 480; conversion EU 200.

Case & worked answers

Pre-assignment: Nusantara CraftWorks — full solution

Part 1: items 1, 2, 4 direct manufacturing product costs; 3, 5, 6, 7 indirect manufacturing overhead; 8, 9 traceable to the order but selling (period) costs; 10 indirect administrative period cost. Prime cost IDR 276m; conversion IDR 240m. High-low: b = IDR 5,000/MH, a = IDR 6m → Y = 6,000,000 + 5,000X; August 750 MH → IDR 9.75m. Headboard: sunk cost IDR 60m; opportunity cost IDR 3m rent; relevant cost A = 18+9+2+3 = 32m, B = 32m — a tie, so qualitative factors decide; the machine's price is irrelevant.

Part 2: POHR = 720m ÷ 24,000 = IDR 30,000/DLH. KR-118: DM 42m + DL 12m + MOH 4.8m = 58.8m; IDR 2.94m per chair. Incremental cost ≈ 2.82m (excludes fixed overhead). COGM = 1,270m; unadjusted COGS 1,210m; underapplied overhead 10m → adjusted COGS 1,220m.

Part 3: EU materials 7,000, conversion 6,900; cost/EU 12,000 + 9,000 = 21,000; ending WIP 3.3m; transferred 142.8m; reconciles to 146.1m.

Closing: unique vs homogeneous output dictates the system; swapping would cause admin overload or mispricing. Semi-custom shelving → operation costing. High-low can split overhead into fixed and variable to set next year's POHR accurately.

Worked examples

Problem 1 **Easy** **Metode high-low**

Biaya pemeliharaan: bulan tersibuk 9.000 jam mesin = Rp52.000; bulan tersepi 4.000 jam = Rp32.000. Pisahkan biaya tetap dan variabel, lalu perkirakan biaya pada 7.000 jam.

1. ```
   b = (Biaya tertinggi − Biaya terendah) ÷ (Aktivitas tertinggi − terendah) = 20.000 ÷ 5.000 = 4 per jam
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2. a = Biaya total − b × aktivitas = 52.000 − 4 × 9.000 = 16.000 (biaya tetap).
3. Y = 16.000 + 4 × 7.000 = 44.000.

Answer: Y = 16.000 + 4X; pada 7.000 jam biayanya Rp44.000.
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Problem 2 Medium **Job-order costing**

Estimasi overhead setahun Rp600.000 dengan dasar 40.000 jam tenaga kerja langsung. Job #7: bahan langsung 8.000, tenaga kerja langsung 6.000 (400 jam), 500 unit. Harga jual = biaya + markup 50%. Berapa harga per unit?

1. `POHR = 600.000 ÷ 40.000 = 15 per jam`
2. Overhead dibebankan = 400 × 15 = 6.000.
3. Total biaya job = 8.000 + 6.000 + 6.000 = 20.000 → per unit = 20.000 ÷ 500 = 40.
4. Harga = 40 × 1,5 = **60**.

Answer: Biaya per unit 40; harga jual 60.

Problem 3 Challenging **Unit ekuivalen (rata-rata tertimbang)**

Departemen pencampuran: unit selesai 16.000; WIP akhir 4.000 unit (bahan 100%, konversi 25%). Total biaya konversi Rp340.000. Hitung unit ekuivalen bahan dan konversi, biaya konversi per unit ekuivalen, dan biaya konversi WIP akhir.

1. Bahan: 16.000 + 4.000 × 100% = **20.000** unit ekuivalen.
2. Konversi: 16.000 + 4.000 × 25% = **17.000** unit ekuivalen.
3. Biaya konversi per UE = 340.000 ÷ 17.000 = **20**.
4. Biaya konversi WIP akhir = 1.000 × 20 = 20.000; yang selesai = 16.000 × 20 = 320.000. Cek: 320.000 + 20.000 = 340.000 ✓

Answer: UE bahan 20.000, UE konversi 17.000; Rp20 per UE; WIP akhir (konversi) Rp20.000.

Practice questions

Try answering first. The answer key is below each question.

1. High-low: 900 h = 10.5m, 500 h = 8.5m. Variable cost per hour...

A. 2,000
B. 5,000
C. 11,667
D. 17,000

Answer: B. 2m ÷ 400 = 5,000.

2. POHR 30,000/DLH, job uses 160 DLH. Applied overhead…

A. 480k
B. 4.8m
C. 12m
D. 48m

Answer: B. 4.8m.

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3. A continuous bottled-tea producer should use…

A. Job-order
B. Process costing
C. None
D. Operation only

Answer: B. Uniform output → process costing.

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4. A machine bought 3 years ago for 60m is, for today's decision…

A. Relevant
B. Opportunity cost
C. Sunk cost — irrelevant
D. Variable

Answer: C. Already spent.

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5. 500 WIP units 20% complete for conversion equal…

A. 20
B. 100
C. 400
D. 500

Answer: B. 100.

Sources

- -P3MD- Week 4 MK02.pdf
- Pre-Assignment MK02 Week 4 Nusantara CraftWorks Case Study.pdf